IDRAAK MOHD KHAN

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Education

Jaypee Institute of Information Technology, Noida **2022 - 2026**
B.Tech. in Electronics and Communication Engineering

Experience

Software Development Engineering Intern - *Amazon* *Jan 2026 - Jun 2026*

Project 1: "Marketplace Expansion Service - End-to-End Backend & AWS Infrastructure"

Delivered an end-to-end Java microservice and AWS infrastructure for marketplace expansion workflows, spanning API design, business logic, data persistence, asynchronous processing, networking, security, observability, and deployment.

- **Designed and built** a **Java Springboot microservice**, implementing batch-capable marketplace expansion APIs backed by **DynamoDB (NoSQL)** and **SQS**.
- **Architected and provisioned the beta and production cloud infrastructure with AWS CDK**, integrating **API Gateway, VPC Link, NLB, and ECS Fargate**, with IAM-based access control.
- **Engineered reliability and operational readiness** through DynamoDB batch operations, **idempotent request processing, retries, and partial-failure handling**, along with CloudWatch dashboards and alarms.
- Developed **unit tests** (JUnit), **end-to-end integration tests,** and configured, modified, and managed **CI/CD pipelines** using Amazon's Pipelines platform to **automate build validation** and **deployment workflows**.
- **Implemented a phased production rollout using Amazon WebLab**, progressively shifting traffic from the legacy service to the new service with **production monitoring and rollback safeguards**.

Project 2: "Selection & Growth Hub Website BFF Modernization - Spring Boot to AWS AppSync"

Modernized the application backend by replacing a Spring Boot BFF with AWS AppSync and redesigning the frontend integration for a production-ready migration.

- **Migrated the Spring Boot BFF to AppSync**, designing the **GraphQL schema, data sources, and resolvers** to route frontend requests to the backend services while replacing the existing REST-based API layer.
- **Migrated UI configuration from AWS AppConfig to ACS**, creating the TypeScript configuration client and integrating offline configuration artifacts into the React application; **redesigned frontend API flows** to retrieve UI configuration through ACS and application data through **AppSync**, removing the dependency on the old BFF
- **Delivered the migration through a phased production rollout using Amazon WebLab**, progressively shifting traffic from the legacy BFF to AppSync (**0%** -> **10%** -> **50%** -> **100%**) with production monitoring and rollback safeguards before retiring the legacy infrastructure.

AI Engineering Intern - *Code Gain Technologies Pvt. Ltd.* *Jul 2025 - Dec 2025*

Project 1: "RAG-Based YouTube Lecture Assistant"

Developed a **full-stack application** using **FastAPI, React, and TypeScript**, with **PostgreSQL** for persistent application data, converting YouTube/uploaded lectures into **RAG-based Q&A, quizzes, summaries, etc.**

- Built a **hybrid RAG pipeline** using **BM25 and Sentence-Transformer embeddings** for lexical and semantic retrieval, integrating relevant lecture context into LLM prompts.
- Implemented **real-time SSE streaming** for Gemini responses with **automatic model fallback and retry handling** for rate limits and transient failures.
- Built **LLM-based assessment and personalization** with automated question generation, short-answer grading, **spaced-repetition flashcards**, and quiz-history-based weakness detection.

Skills

Languages & Concepts: Java, Python, C++, Data Structures and Algorithms, OOP, SOLID, DBMS, Git, Distributed Systems, Agile/Scrum/Sprint.
Development: Spring Boot, REST APIs, GraphQL, Microservices, PostgreSQL, NoSQL.
AI & Retrieval: RAG, LLMs, SentenceTransformers, Embeddings, Semantic Search, Vector Search, BM25.
AWS, Cloud, and DevOps: AWS AppSync, API Gateway, NLB, VPC, ECS Fargate, DynamoDB, SQS, IAM, AWS CDK, CloudWatch, Docker, CI/CD pipelines, Automated Testing, Integration Testing, End-to-End Testing.
Frontend: React, TypeScript.